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ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech (Aerospace Engineering)

Semester: V

Name of the Paper: Aerodynamics - II

Paper Code: **TAS-502**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks
- (v) Gas Tables are allowed during the examination.

Q1 (20 marks)		CO1
(a)	Define Stagnation Properties. Using the energy equation, derive the relationship between static and stagnation properties for an isentropic flow. For a supersonic wind tunnel nozzle with $T_0 = 500$ K and $p_0 = 10$ atm, if the exit pressure is $p = 1$ atm, calculate the exit temperature and density.	
(b)	Derive the Prandtl Relation and hence obtain the equations for estimating changes in flow properties across a normal shock wave.	
(c)	Explain the difference in methodology for measuring flow velocity using a Pitot tube in subsonic and supersonic flows. In a supersonic wind tunnel test section, a Pitot tube records a pressure of 1.13 atm while the static pressure measured from the sidewall is 0.1 atm. Determine the Mach number of the flow in the test section.	
Q2 (20 marks)		CO2
(a)	A Mach 4 airflow at 1 atm is slowed to subsonic speed using different shock systems. Compare the total pressure losses for: (a) A single normal shock. (b) A two-step oblique system: first oblique shock with $\theta_1 = 25.3^\circ$ and second oblique shock wave with $\theta_2 = 20^\circ$, followed by a normal shock.	
(b)	A wedge with half-angle 30.2° is placed in a Mach 3.5 freestream at $p_{inf} = 0.5$ atm. A Pitot tube is located behind the shock. Calculate the Pitot pressure.	
(c)	A compression corner produces an oblique shock with flow deflection angle of 18.2° and shock wave subsequently impinges on a flat wall and is further reflected. The upstream flow has $M_1 = 3.2$, $p_1 = 1$ atm, $T_1 = 288$ K. Find M_3 , p_3 , T_3 behind the reflected shock and the angle it makes with the flat wall.	

Q3 (20 marks)		CO3
(a)	An infinitely thin flat plate is exposed to a Mach 2 airflow at an angle of attack of 30° . Compute the lift and wave drag coefficients for the plate.	
(b)	A supersonic flow at $M_1 = 2$, $T_1 = 285 \text{ K}$, $p_1 = 1 \text{ atm}$ is turned upward through a compression corner with $\theta = 32^\circ$, and then expanded by the same angle so that the exit flow is aligned with the freestream direction. Determine M_3 , p_3 , T_3 downstream of the expansion corner.	
(c)	A diamond-wedge airfoil with a half-angle $\varepsilon = 5^\circ$ is placed at an angle of attack $\alpha = 10^\circ$ in a Mach 3 freestream flow. Evaluate the lift coefficient and wave-drag coefficient for this configuration.	
Q4 (20 marks)		CO5
(a)	Discuss small perturbation theory in brief? Derive the two-dimensional compressible flow equation based on the small-disturbance (perturbation) potential formulation.	
(b)	Explain the Prandtl–Glauert compressibility correction and derive the corresponding mathematical expression that relates the compressible and incompressible lift coefficients.	
(c)	Estimate the approximate lift coefficient of a thin airfoil placed at an angle of attack of 5° in a Mach 0.7 flow, using an appropriate theoretical analytical relation.	
Q5 (20 marks)		CO4
(a)	Discuss how supercritical airfoils are specially designed to delay the onset of drag divergence and improve aerodynamic efficiency at transonic speeds.	
(b)	Describe the performance characteristics of swept wings in transonic and supersonic regimes. Support your explanation with neat and labeled diagrams.	
(c)	Discuss in detail the flow characteristics and aerodynamic properties associated with hypersonic flow.	